JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2025 24th January Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Chemistry	25 questions
Mathematics	25 questions
Physics	25 questions
Answer key	Included at the end

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QR codes open the DekhoExams website.

Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

For a reaction, $N_2O_{5(g)} \rightarrow 2NO_{2(g)} + \frac{1}{2}O_{2(g)}$ in a constant volume container, no products were present initially. The final pressure of the system when 50% of reaction gets completed is

(A) $5/2$ times of initial pressure

(B) $7/2$ times of initial pressure

(C) $7/4$ times of initial pressure

(D) 5 times of initial pressure

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Q2. MCQ (Single Correct Answer)

The large difference between the melting and boiling points of oxygen and sulphur may be explained on the basis of

(A) Atomicity

(B) Electron gain enthalpy

(C) Atomic size

(D) Electronegativity

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Q3. MCQ (Single Correct Answer)

Given below are two statements I and II.

Statement I : Dumas method is used for estimation of "Nitrogen" in an organic compound.

Statement II : Dumas method involves the formation of ammonium sulphate by heating the organic compound with conc H_2SO_4.

In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is true but Statement II is false

(B) Statement I is false but Statement II is true

(C) Both Statement I and Statement II are true

(D) Both Statement I and Statement II are false

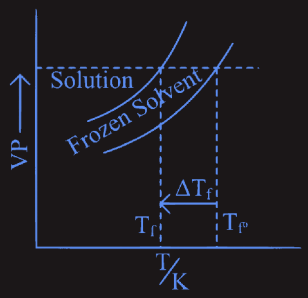
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Q4. MCQ (Single Correct Answer)

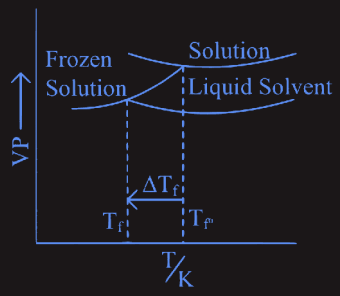
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Consider the given plots of vapour pressure (VP) vs temperature(T/K). Which amongst the following options is correct graphical representation showing ΔT_f, depression in the freezing point of a solvent in a solution?

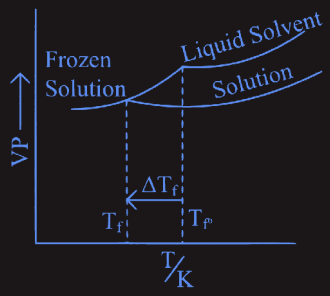
(A)



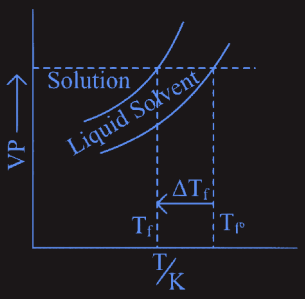
(B)



(C)



(D)



Q5. MCQ (Single Correct Answer)

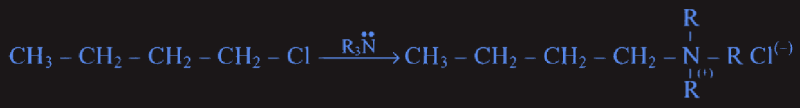
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Given below are two statements:

Statement I: The conversion proceeds well in the less polar medium.

$$CH_3 - CH_2 - CH_2 - CH_2 - Cl \xrightarrow{HO^-} CH_3 - CH_2 - CH_2 - CH_2 - OH + Cl^{(-)}$$

Statement II: The conversion proceeds well in the more polar medium.



In the light of the above statements, choose the correct answer from the options given below

(A) Statement I is false but Statement II is true

(B) Statement I is true but Statement II is false

(C) Both Statement I and Statement II are false

(D) Both Statement I and Statement II are true

Q6. MCQ (Single Correct Answer)

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Preparation of potassium permanganate from MnO_2 involves two step process in which the 1st step is a reaction with KOH and KNO_3 to produce

(A) K_3MnO_4

(B) $K_4[Mn(OH)_6]$

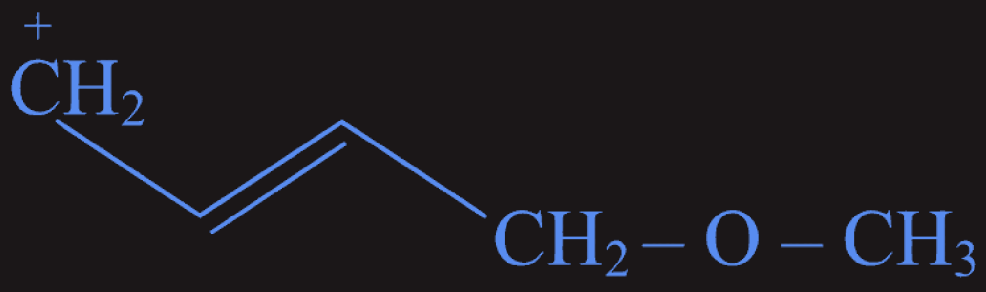
(C) K_2MnO_4

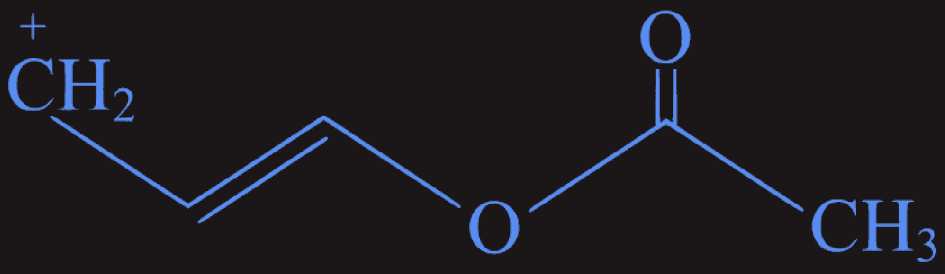
(D) $KMnO_4$

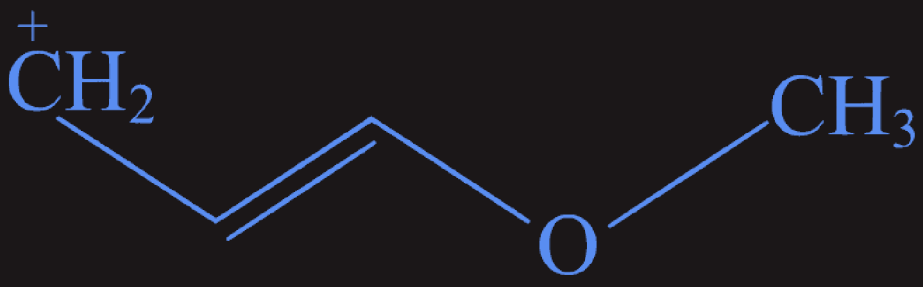
Q7. MCQ (Single Correct Answer)

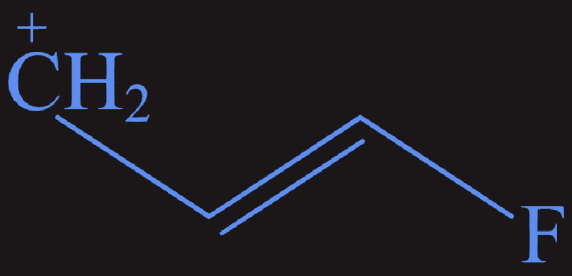
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Which one of the carbocations from the following is most stable?

(A) 

(B) 

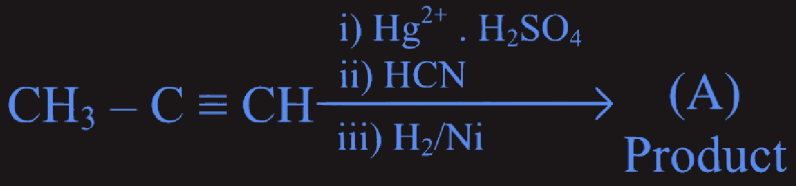
(C) 

(D) 

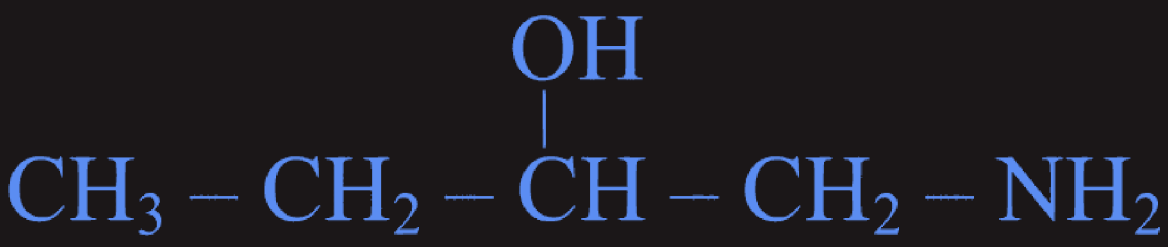
Q8. MCQ (Single Correct Answer)

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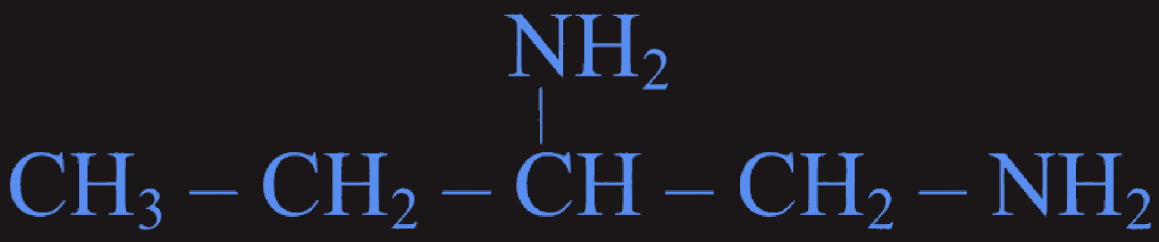
The product (A) formed in the following reaction sequence is



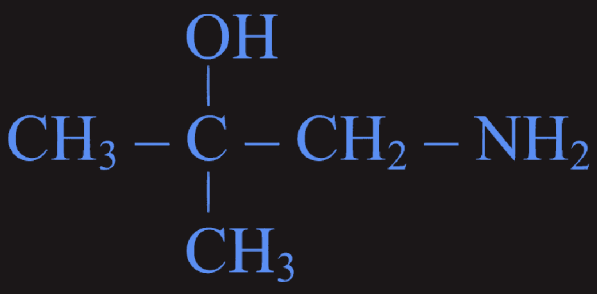
(A)



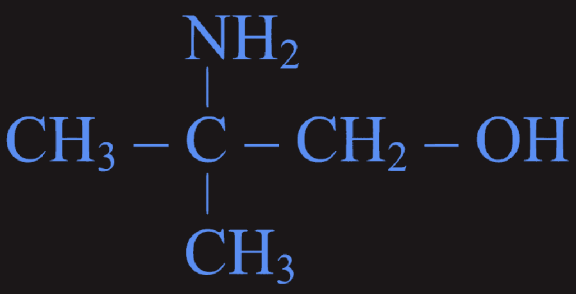
(B)



(C)



(D)



Q9. MCQ (Single Correct Answer)

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K_{sp} for $Cr(OH)_3$ is 1.6×10^{-30}. What is the molar solubility of this salt in water?

(A) $\sqrt[5]{1.8 \times 10^{-30}}$

(B) $\frac{1.8\times10^{-30}}{27}$

(C) $\sqrt[4]{\frac{1.6\times10^{-30}}{27}}$

(D) $\sqrt[2]{1.6 \times 10^{-30}}$

Q10. MCQ (Single Correct Answer)

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Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z-direction] ?

A. $2\mathrm{p}_\mathrm{z}$ and $2\mathrm{p}_\mathrm{x}$

B. 2 s and $2\mathrm{p}_\mathrm{x}$

C. $3d_{xy}$ and $3d_{x^2-y^2}$

D. 2 s and $2\mathrm{p}_\mathrm{z}$

E. $2p_z$ and $3d_{x^2-y^2}$

Choose the correct answer from the options given below:

(A) D Only (B) E Only (C) C and D Only (D) A and B Only

Q11. MCQ (Single Correct Answer)

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Which of the following statements are NOT true about the periodic table?

A. The properties of elements are function of atomic weights.

B. The properties of elements are function of atomic numbers.

C. Elements having similar outer electronic configurations are arranged in same period.

D. An element's location reflects the quantum numbers of the last filled orbital.

E. The number of elements in a period is same as the number of atomic orbitals available in energy level that is being filled.

Choose the correct answer from the options given below:

(A) A, C and E Only (B) A and E Only (C) B, C and E Only (D) D and E Only

Q12. MCQ (Single Correct Answer)

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One mole of the octahedral complex compound $\mathrm{Co(NH_3)_5Cl_3}$ gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with excess of $\mathrm{AgNO_3}$ solution to yield two moles of $\mathrm{AgCl_{(s)}}$. The structure of the complex is:

(A) $\mathrm{[Co(NH_3)_5Cl]Cl_2}$

(B) $\mathrm{[Co(NH_3)_4Cl]\cdot Cl_2\cdot NH_3}$

(C) $\mathrm{[Co(NH_3)_3Cl_3]\cdot 2NH_3}$

(D) $\mathrm{[Co(NH_3)_4Cl_2]\cdot Cl\cdot NH_3}$

Q13. MCQ (Single Correct Answer)

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Which of the following statement is true with respect to H_2O, NH_3 and CH_4 ?

A. The central atoms of all the molecules are sp^3 hybridized.

B. The $H-O-H, H-N-H$ and $H-C-H$ angles in the above molecules are $104.5^\circ, 107.5^\circ$ and 109.5°, respectively.

C. The increasing order of dipole moment is $CH_4 < NH_3 < H_2O$.

D. Both H_2O and NH_3 are Lewis acids and CH_4 is a Lewis base.

E. A solution of NH_3 in H_2O is basic. In this solution NH_3 and H_2O act as Lowry-Bronsted acid and base respectively.

Choose the correct answer from the options given below:

(A) A, D and E Only

(B) C, D and E Only

(C) A, B and C Only

(D) A, B, C and E Only

Q14. MCQ (Single Correct Answer)

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Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option.

(A) Both ΔH and ΔS are (-ve)

(B) ΔH is $(+ve)$ but ΔS is (-ve)

(C) ΔH is $(-ve)$ but ΔS is (+ve)

(D) Both ΔH and ΔS are (+ve)

Q15. MCQ (Single Correct Answer)

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Which of the following arrangements with respect to their reactivity in nucleophilic addition reaction is correct?

(A) acetophenone < benzaldehyde < p-tolualdehyde < p-nitrobenzaldehyde

(B) acetophenone < p-tolualdehyde < benzaldehyde < p-nitrobenzaldehyde

(C) p-nitrobenzaldehyde < benzaldehyde < p-tolualdehyde < acetophenone

(D) benzaldehyde < acetophenone < p-nitrobenzaldehyde < p-tolualdehyde

Q16. MCQ (Single Correct Answer)

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The carbohydrate "Ribose" present in DNA, is

A. A pentose sugar

B. present in pyranose form

C. in "D" configuration

D. a reducing sugar, when free

E. in α-anomeric form

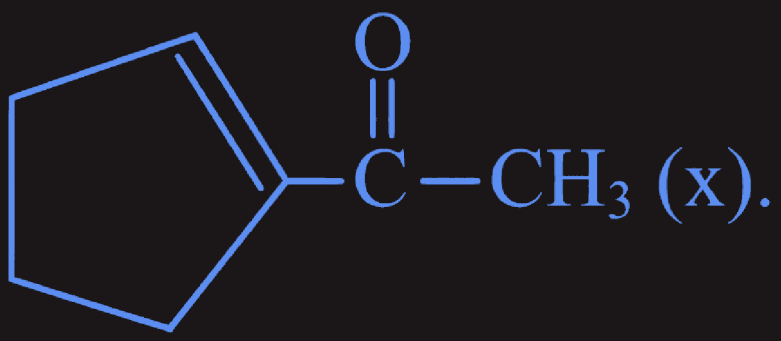
Choose the correct answer from the options given below:

(A) A, D and E Only **(B)** B, D and E Only **(C)** A, C and D Only **(D)** A, B and E Only

Q17. MCQ (Single Correct Answer)

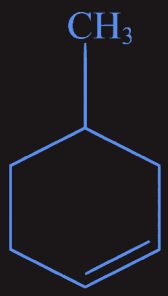
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Aman has been asked to synthesise the molecule



He thought of preparing the molecule using an aldol condensation reaction. He found a few cyclic alkenes in his laboratory. He thought of performing ozonolysis reaction on alkene to produce a dicarbonyl compound followed by aldol reaction to prepare " x ". Predict the suitable alkene that can lead to the formation of " x ".

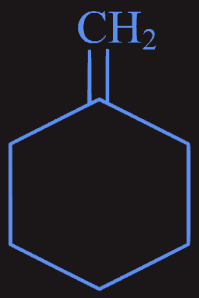
(A)



(B)



(C)



(D)



Q18. MCQ (Single Correct Answer)

For the given cell

$Fe^{2+}(aq) + Ag^{+}_{(aq)} \rightarrow Fe^{3+}(aq) + Ag_{(s)}$

The standard cell potential of the above reaction is Given:

$Ag^{+} + e^{-} \rightarrow Ag \quad E^{\theta} = xV$

$Fe^{2+} + 2e^{-} \rightarrow Fe \quad E^{\theta} = yV$

$Fe^{3+} + 3e^{-} \rightarrow Fe \quad E^{\theta} = zV$

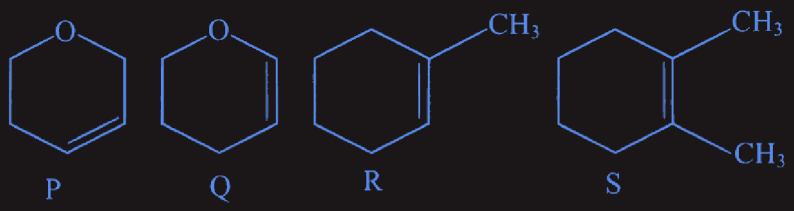
(A) $x + 2y - 3z$ (B) $x + 2y$ (C) $y - 2x$ (D) $x + y - z$

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Q19. MCQ (Single Correct Answer)

Following are the four molecules "P", "Q", "R" and "S".

Which one among the four molecules will react with $H - Br_{(aq)}$ at the fastest rate?



(A) Q (B) S (C) P (D) R

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Q20. MCQ (Single Correct Answer)

Which of the following ions is the strongest oxidizing agent? (Atomic Number of $Ce = 58, Eu = 63, Tb = 65, Lu = 71$)

(A) Lu^{3+} (B) Eu^{2+} (C) Tb^{4+} (D) Ce^{3+}

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Q21. Numerical

$37.8\ g\ N_2O_5$ was taken in a 1 L reaction vessel and allowed to undergo the following reaction at 500 K

$2\ N_2O_{5(g)} \rightleftharpoons 2\ N_2O_{4(g)} + O_{2(g)}$

The total pressure at equilibrium was found to be 18.65 bar.

Then, $Kp =$ _____ $\times 10^{-2}$ [nearest integer]

Assume N_2O_5 to behave ideally under these conditions.

Given: $R = 0.082$ bar $Lmol^{-1}\ K^{-1}$

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Q22. Numerical

Among the following cations, the number of cations which will give characteristic precipitate in their identification tests with $K_4[Fe(CN)_6]$ is _____.

$Cu^{2+}, Fe^{3+}, Ba^{2+}, Ca^{2+}, NH_4^{+}, Mg^{2+}, Zn^{2+}$

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Q23. Numerical

Consider the following reaction occurring in the blast furnace:

$Fe_3O_{4(s)} + 4CO_{(g)} \rightarrow 3Fe_{(l)} + 4CO_{2(g)}$

' x ' kg of iron is produced when 2.32×10^3 kgFe_3O_4 and 2.8×10^2 kgCO are brought together in the furnace. The value of ' x ' is _____ . (nearest integer)

{Given: molar mass of $Fe_3O_4 = 232$ g mol^{-1}

molar mass of $CO = 28$ g mol^{-1}

molar mass of $Fe = 56$ g mol^{-1}}

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Q24. Numerical

Xg of benzoic acid on reaction with aq $NaHCO_3$ released CO_2 that occupied 11.2 L volume at STP.

X is _____ g.

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Q25. Numerical

Standard entropies of X_2, Y_2 and XY_5 are 70, 50 and 110 J K^{-1} mol^{-1} respectively. The temperature in Kelvin at which the reaction

$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5 \Delta H^{\ominus} = -35$ kJ mol^{-1}

will be at equilibrium is _____ (Nearest integer)

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

$\lim\limits_{x \to 0} \operatorname{cosec} x \left(\sqrt{2\cos^2 x + 3\cos x} - \sqrt{\cos^2 x + \sin x + 4}\right)$ is:

(A) $\frac{1}{\sqrt{15}}$ (B) $\frac{1}{2\sqrt{5}}$ (C) 0 (D) $-\frac{1}{2\sqrt{5}}$

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Q27. MCQ (Single Correct Answer)

Let $f : \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ be a function such that $f(x) - 6f\left(\frac{1}{x}\right) = \frac{35}{3x} - \frac{5}{2}$. If the $\lim\limits_{x \to 0}\left(\frac{1}{\alpha x} + f(x)\right) = \beta; \alpha, \beta \in \mathbb{R}$, then $\alpha + 2\beta$ is equal to

(A) 6 (B) 5 (C) 3 (D) 4

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Q28. MCQ (Single Correct Answer)

The product of all the rational roots of the equation $\left(x^2 - 9x + 11\right)^2 - (x-4)(x-5) = 3$, is equal to

(A) 7 (B) 21 (C) 28 (D) 14

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Q29. MCQ (Single Correct Answer)

For some $\mathrm{n} \neq 10$, let the coefficients of the 5 th, 6 th and 7 th terms in the binomial expansion of $(1+\mathrm{x})^{\mathrm{n}+4}$ be in A.P. Then the largest coefficient in the expansion of $(1+\mathrm{x})^{\mathrm{n}+4}$ is:

(A) 10 (B) 35 (C) 70 (D) 20

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Q30. MCQ (Single Correct Answer)

Let $f(x)=\frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right)+f\left(\frac{2}{15}\right)+\ldots+f\left(\frac{59}{15}\right)\right)$ is equal to

(A) 108 (B) 92 (C) 118 (D) 102

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Q31. MCQ (Single Correct Answer)

If $I(m,n)=\int_0^1 x^{m-1}(1-x)^{n-1}dx, m,n>0$, then $I(9,14)+I(10,13)$ is

(A) $I(9,1)$ (B) $I(1,13)$ (C) $\mathrm{I}(19,27)$ (D) $\mathrm{I}(9,13)$

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Q32. MCQ (Single Correct Answer)

If α and β are the roots of the equation $2z^2-3z-2i=0$, where $i=\sqrt{-1}$, then $16\cdot \mathrm{Re}\left(\frac{\alpha^{19}+\beta^{19}+\alpha^{11}+\beta^{11}}{\alpha^{15}+\beta^{15}}\right)\cdot \mathrm{Im}\left(\frac{\alpha^{19}+\beta^{19}+\alpha^{11}+\beta^{11}}{\alpha^{15}+\beta^{15}}\right)$ is equal to

(A) 441 (B) 312 (C) 409 (D) 398

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Q33. MCQ (Single Correct Answer)

Let the product of the focal distances of the point $\left(\sqrt{3},\frac{1}{2}\right)$ on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1,(a>b)$, be $\frac{7}{4}$. Then the absolute difference of the eccentricities of two such ellipses is

(A) $\frac{1-2\sqrt{2}}{\sqrt{3}}$ (B) $\frac{1-\sqrt{3}}{\sqrt{2}}$ (C) $\frac{3-2\sqrt{2}}{2\sqrt{3}}$ (D) $\frac{3-2\sqrt{2}}{3\sqrt{2}}$

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Q34. MCQ (Single Correct Answer)

Let $\mathrm{y}=\mathrm{y}(\mathrm{x})$ be the solution of the differential equation $\left(xy-5x^2\sqrt{1+x^2}\right)dx+\left(1+x^2\right)dy=0, y(0)=0$. Then $y(\sqrt{3})$ is equal to

(A) $\frac{5\sqrt{3}}{2}$ (B) $\sqrt{\frac{15}{2}}$ (C) $\sqrt{\frac{14}{3}}$ (D) $2\sqrt{2}$

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Q35. MCQ (Single Correct Answer)

Let in a $\triangle ABC$, the length of the side AC be 6 , the vertex B be $(1,2,3)$ and the vertices A, C lie on the line $\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$. Then the area (in sq. units) of $\triangle ABC$ is:

(A) 42 (B) 17 (C) 56 (D) 21

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Q36. MCQ (Single Correct Answer)

Let the line passing through the point $(-1, 2, 1)$ and parallel to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ intersect the line $\frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1}$ at the point P. Then the distance of P from the point $Q(4, -5, 1)$ is

(A) $5\sqrt{6}$ (B) 5 (C) $5\sqrt{5}$ (D) 10

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Q37. MCQ (Single Correct Answer)

A and B alternately throw a pair of dice. A wins if he throws a sum of 5 before B throws a sum of 8 , and B wins if he throws a sum of 8 before A throws a sum of 5 . The probability, that A wins if A makes the first throw, is

(A) $\frac{8}{19}$ (B) $\frac{9}{19}$ (C) $\frac{8}{17}$ (D) $\frac{9}{17}$

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Q38. MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{c}$ is coplanar with $\vec{a}$ and $\vec{b}$. If the vector $\vec{c}$ is perpendicular to $\vec{b}$ and $\vec{a} \cdot \vec{c} = 5$, then $|\vec{c}|$ is equal to

(A) $\sqrt{\frac{11}{6}}$ (B) $\frac{1}{3\sqrt{2}}$ (C) 18 (D) 16

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Q39. MCQ (Single Correct Answer)

Consider the region $R = \left\{(x, y) : x \leq y \leq 9 - \frac{11}{3}x^2, x \geq 0\right\}$. The area of the largest rectangle of sides parallel to the coordinate axes and inscribed in R , is:

(A) $\frac{821}{123}$ (B) $\frac{567}{121}$ (C) $\frac{730}{119}$ (D) $\frac{625}{111}$

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Q40. MCQ (Single Correct Answer)

The area of the region $\left\{(x, y) : x^2 + 4x + 2 \leq y \leq |x + 2|\right\}$ is equal to

(A) 7 (B) 24/5 (C) 20/3 (D) 5

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Q41. MCQ (Single Correct Answer)

Let $S_n = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \ldots$ upto n terms. If the sum of the first six terms of an A.P. with first term -p and common difference p is $\sqrt{2026\, S_{2025}}$, then the absolute difference between 20th and 15th terms of the A.P. is

(A) 20 (B) 45 (C) 90 (D) 25

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Q42. MCQ (Single Correct Answer)

For a statistical data $x_1, x_2, \ldots, x_{10}$ of 10 values, a student obtained the mean as 5.5 and $\sum_{i=1}^{10} x_i^2 = 371$. He later found that he had noted two values in the data incorrectly as 4 and 5 , instead of the correct values 6 and 8 , respectively. The variance of the corrected data is

(A) 5 (B) 7 (C) 9 (D) 4

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Q43. MCQ (Single Correct Answer)

Let the lines $3x-4y-\alpha=0, 8x-11y-33=0$, and $2x-3y+\lambda=0$ be concurrent. If the image of the point $(1,2)$ in the line $2x-3y+\lambda=0$ is $\left(\frac{57}{13}, \frac{-40}{13}\right)$, then $|\alpha\lambda|$ is equal to

(A) 91 (B) 113 (C) 101 (D) 84

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Q44. MCQ (Single Correct Answer)

If the system of equations

$2x-y+z=4$
$5x+\lambda y+3z=12$
$100x-47y+\mu z=212$

has infinitely many solutions, then $\mu-2\lambda$ is equal to

(A) 56 (B) 59 (C) 57 (D) 55

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Q45. MCQ (Single Correct Answer)

Let circle C be the image of $x^2+y^2-2x+4y-4=0$ in the line $2x-3y+5=0$ and A be the point on C such that OA is parallel to x-axis and A lies on the right hand side of the centre O of C. If $B(\alpha,\beta)$, with $\beta<4$, lies on C such that the length of the arc AB is $(1/6)^{\text{th}}$ of the perimeter of C, then $\beta-\sqrt{3}\alpha$ is equal to

(A) $4-\sqrt{3}$ (B) 3 (C) 4 (D) $3+\sqrt{3}$

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Q46. Numerical

Let A be a 3×3 matrix such that $\mathrm{X}^{\mathrm{T}}\mathrm{AX}=\mathrm{O}$ for all nonzero 3×1 matrices $X=\begin{bmatrix}x\\y\\z\end{bmatrix}$. If $\mathrm{A}\begin{bmatrix}1\\1\\1\end{bmatrix}=\begin{bmatrix}1\\4\\-5\end{bmatrix}$, $\mathrm{A}\begin{bmatrix}1\\2\\1\end{bmatrix}=\begin{bmatrix}0\\4\\-8\end{bmatrix}$, and $\det(\operatorname{adj}(2(\mathrm{A}+\mathrm{I})))=2^{\alpha}3^{\beta}5^{\gamma}, \alpha,\beta,\gamma\in N$, then $\alpha^2+\beta^2+\gamma^2$ is

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Q47. Numerical

Let f be a differentiable function such that $2(x+2)^2f(x)-3(x+2)^2=10\int_0^x(t+2)f(t)dt, x\geq0$. Then $f(2)$ is equal to _____ .

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Q48. Numerical

Let $S=\{p_1,p_2\ldots,p_{10}\}$ be the set of first ten prime numbers. Let $A=S\cup P$, where P is the set of all possible products of distinct elements of S. Then the number of all ordered pairs $(x,y), x\in S$, $y\in A$, such that x divides y, is _____ .

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Q49. Numerical

If for some $\alpha, \beta; \alpha \leq \beta, \alpha + \beta = 8$ and $\sec^2\left(\tan^{-1}\alpha\right) + \operatorname{cosec}^2\left(\cot^{-1}\beta\right) = 36$, then $\alpha^2 + \beta$ is _____

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Q50. Numerical

The number of 3 -digit numbers, that are divisible by 2 and 3 , but not divisible by 4 and 9 , is _____.

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

An object of mass ' m ' is projected from origin in a vertical xy plane at an angle 45° with the $\mathrm{x}-$ axis with an initial velocity v_0. The magnitude and direction of the angular momentum of the object with respect to origin, when it reaches at the maximum height, will be [g is acceleration due to gravity]

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(A) $\frac{mv_o^3}{2\sqrt{2}g}$ along negative z-axis

(B) $\frac{mv_o^3}{2\sqrt{2}g}$ along positive z-axis

(C) $\frac{mv_o^3}{4\sqrt{2}g}$ along positive z-axis

(D) $\frac{mv_o^3}{4\sqrt{2}g}$ along negative z-axis

Q52. MCQ (Single Correct Answer)

The Young's double slit interference experiment is performed using light consisting of 480 nm and 600 nm wavelengths to form interference patterns. The least number of the bright fringes of 480 nm light that are required for the first coincidence with the bright fringes formed by 600 nm light is

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(A) 8 (B) 4 (C) 5 (D) 6

Q53. MCQ (Single Correct Answer)

What is the relative decrease in focal length of a lens for an increase in optical power by 0.1 D from 2.5D ? ['D' stands for dioptre]

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(A) 0.04 (B) 0.1 (C) 0.40 (D) 0.01

Q54. MCQ (Single Correct Answer)

A car of mass ' m ' moves on a banked road having radius ' r ' and banking angle θ. To avoid slipping from banked road, the maximum permissible speed of the car is v_0. The coefficient of friction μ between the wheels of the car and the banked road is

(A) $\mu = \frac{v_0^2 + rg\tan\theta}{rg + v_0^2\tan\theta}$

(B) $\mu = \frac{v_0^2 - rg\tan\theta}{\mathrm{rg} - \mathrm{v}_o^2\tan\theta}$

(C) $\mu = \frac{v_0^2 - rg\tan\theta}{rg + v_0^2\tan\theta}$

(D) $\mu = \frac{v_o^2 + rg\tan\theta}{rg - v_o^2\tan\theta}$

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Q55. MCQ (Single Correct Answer)

During the transition of electron from state A to state C of a Bohr atom, the wavelength of emitted radiation is $2000\overset{o}{A}$ and it becomes $6000\overset{o}{A}$ when the electron jumps from state B to state C. Then the wavelength of the radiation emitted during the transition of electrons from state A to state B is

(A) $4000\overset{o}{A}$ (B) $6000\overset{o}{A}$ (C) $2000\overset{o}{A}$ (D) $3000\overset{o}{A}$

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Q56. MCQ (Single Correct Answer)

A satellite is launched into a circular orbit of radius ' R ' around the earth. A second satellite is launched into an orbit of radius 1.03 R . The time period of revolution of the second satellite is larger than the first one approximately by

(A) 3% (B) 2.5% (C) 4.5% (D) 9%

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Q57. MCQ (Single Correct Answer)

A particle is executing simple harmonic motion with time period 2 s and amplitude 1 cm . If D and d are the total distance and displacement covered by the particle in 12.5 s , then $\frac{\mathrm{D}}{\mathrm{d}}$ is

(A) 10 (B) $\frac{16}{5}$ (C) 25 (D) $\frac{15}{4}$

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Q58. MCQ (Single Correct Answer)

An air bubble of radius 0.1 cm lies at a depth of 20 cm below the free surface of a liquid of density $1000\ \mathrm{kg/m^3}$. If the pressure inside the bubble is $2100\ \mathrm{N/m^2}$ greater than the atmospheric pressure, then the surface tension of the liquid in SI unit is (use $g = 10\ \mathrm{m/s^2}$)

(A) 0.02 (B) 0.05 (C) 0.25 (D) 0.1

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Q59. MCQ (Single Correct Answer)

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An ideal gas goes from an initial state to final state. During the process, the pressure of gas increases linearly with temperature.

A. The work done by gas during the process is zero.

B. The heat added to gas is different from change in its internal energy.

C. The volume of the gas is increased.

D. The internal energy of the gas is increased.

E. The process is isochoric (constant volume process)

Choose the correct answer from the options given below:

(A) A, C Only (B) A, D, E Only (C) E Only (D) A, B, C, D Only

Q60. MCQ (Single Correct Answer)

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Consider a parallel plate capacitor of area A (of each plate) and separation ' d ' between the plates. If E is the electric field and ε_0 is the permittivity of free space between the plates, then potential energy stored in the capacitor is

(A) $\varepsilon_0 \mathrm{E}^2 \mathrm{Ad}$ (B) $\frac{3}{4} \varepsilon_0 \mathrm{E}^2 \mathrm{Ad}$

(C) $\frac{1}{4} \varepsilon_0 \mathrm{E}^2 \mathrm{Ad}$ (D) $\frac{1}{2} \varepsilon_0 \mathrm{E}^2 \mathrm{Ad}$

Q61. MCQ (Single Correct Answer)

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A force $\mathrm{F}=\alpha+\beta \mathrm{x}^2$ acts on an object in the x -direction. The work done by the force is 5 J when the object is displaced by 1 m . If the constant $\alpha=1 \mathrm{~N}$ then β will be

(A) $15 \mathrm{~N} / \mathrm{m}^2$ (B) $10 \mathrm{~N} / \mathrm{m}^2$ (C) $12 \mathrm{~N} / \mathrm{m}^2$ (D) $8 \mathrm{~N} / \mathrm{m}^2$

Q62. MCQ (Single Correct Answer)

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Consider the following statements :

A. The junction area of solar cell is made very narrow compared to a photo diode.

B. Solar cells are not connected with any external bias.

C. LED is made of lightly doped p-n junction.

D. Increase of forward current results in continuous increase of LED light intensity.

E. LEDs have to be connected in forward bias for emission of light.

Choose the correct answer from the options given below :

(A) A, C, E Only (B) B, E Only (C) A, C Only (D) B, D, E Only

Q63. MCQ (Single Correct Answer)

A parallel plate capacitor was made with two rectangular plates, each with a length of $l = 3\ \mathrm{cm}$ and breadth of $\mathrm{b} = 1\ \mathrm{cm}$. The distance between the plates is $3\mu\ \mathrm{m}$. Out of the following, which are the ways to increase the capacitance by a factor of 10 ?

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A. $l = 30\ \mathrm{cm},\ \mathrm{b} = 1\ \mathrm{cm},\ \mathrm{d} = 1\mu\ \mathrm{m}$

B. $l = 3\ \mathrm{cm},\ \mathrm{b} = 1\ \mathrm{cm},\ \mathrm{d} = 30\mu\ \mathrm{m}$

C. $l = 6\ \mathrm{cm},\ \mathrm{b} = 5\ \mathrm{cm},\ \mathrm{d} = 3\mu\ \mathrm{m}$

D. $l = 1\ \mathrm{cm},\ \mathrm{b} = 1\ \mathrm{cm},\ \mathrm{d} = 10\mu\ \mathrm{m}$

E. $l = 5\ \mathrm{cm},\ \mathrm{b} = 2\ \mathrm{cm},\ \mathrm{d} = 1\mu\ \mathrm{m}$

Choose the correct answer from the options given below:

(A) C only (B) A only (C) B and D only (D) C and E only

Q64. MCQ (Single Correct Answer)

A thin plano convex lens made of glass of refractive index 1.5 is immersed in a liquid of refractive index 1.2. When the plane side of the lens is silver coated for complete reflection, the lens immersed in the liquid behaves like a concave mirror of focal length 0.2 m . The radius of curvature of the curved surface of the lens is

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(A) 0.15 m (B) 0.10 m (C) 0.25 m (D) 0.20 m

Q65. MCQ (Single Correct Answer)

A plano-convex lens having radius of curvature of first surface 2 cm exhibits focal length of f_1 in air. Another plano-convex lens with first surface radius of curvature 3 cm has focal length of f_2 when it is immersed in a liquid of refractive index 1.2. If both the lenses are made of same glass of refractive index 1.5 , the ratio of f_1 and f_2 will be

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(A) $2:3$ (B) $1:3$ (C) $1:2$ (D) $3:5$

Q66. MCQ (Single Correct Answer)

An electron of mass ' m ' with an initial velocity $\vec{\mathrm{v}} = \mathrm{v}_0\hat{i}\ (\mathrm{v}_0 > 0)$ enters an electric field $\vec{\mathrm{E}} = -\mathrm{E}_\mathrm{o}\hat{\mathrm{k}}$. If the initial de Broglie wavelength is λ_0, the value after time t would be

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(A) $\frac{\lambda_o}{\sqrt{1-\frac{\mathrm{e}^2\mathrm{E}_\mathrm{o}^2\mathrm{t}^2}{\mathrm{m}^2\mathrm{v}_\mathrm{o}^2}}}$

(B) λ_0

(C) $\frac{\lambda_o}{\sqrt{1+\frac{\mathrm{e}^2\mathrm{E}_\mathrm{o}^2\mathrm{t}^2}{\mathrm{m}^2 v_o^2}}}$

(D) $\lambda_\mathrm{o}\sqrt{1+\frac{\mathrm{e}^2\mathrm{E}_\mathrm{o}^2\mathrm{t}^2}{\mathrm{m}^2\mathrm{v}_\mathrm{o}^2}}$

Q67. MCQ (Single Correct Answer)

The amount of work done to break a big water drop of radius ' R ' into 27 small drops of equal radius is 10 J . The work done required to break the same big drop into 64 small drops of equal radius will be

(A) 20 J (B) 10 J (C) 5 J (D) 15 J

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Q68. MCQ (Single Correct Answer)

A uniform solid cylinder of mass ' m ' and radius ' r ' rolls along an inclined rough plane of inclination 45°. If it starts to roll from rest from the top of the plane then the linear acceleration of the cylinder's axis will be

(A) $\sqrt{2}\,\mathrm{g}$ (B) $\frac{1}{\sqrt{2}}\,\mathrm{g}$ (C) $\frac{1}{3\sqrt{2}}\,\mathrm{g}$ (D) $\frac{\sqrt{2}g}{3}$

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Q69. MCQ (Single Correct Answer)

An alternating current is given by $\mathrm{I}=\mathrm{I}_{\mathrm{A}} \sin \omega \mathrm{t}+\mathrm{I}_{\mathrm{B}} \cos \omega \mathrm{t}$. The r.m.s current will be

(A) $\frac{\sqrt{\mathrm{I}_{\mathrm{A}}^2+\mathrm{I}_{\mathrm{B}}^2}}{2}$ (B) $\sqrt{\frac{\mathrm{I}_{\mathrm{A}}^2+\mathrm{I}_{\mathrm{B}}^2}{2}}$

(C) $\frac{\left|\mathrm{I}_{\mathrm{A}}+\mathrm{I}_{\mathrm{B}}\right|}{\sqrt{2}}$ (D) $\sqrt{\mathrm{I}_{\mathrm{A}}^2+\mathrm{I}_{\mathrm{B}}^2}$

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Q70. MCQ (Single Correct Answer)

For an experimental expression $y=\frac{32.3 \times 1125}{27.4}$, where all the digits are significant. Then to report the value of y we should write

(A) $y=1326.186$ (B) $y=1326.2$ (C) $y=1326.19$ (D) $y=1330$

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Q71. Numerical

A wire of resistance 9Ω is bent to form an equilateral triangle. Then the equivalent resistance across any two vertices will be _____ ohm.

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Q72. Numerical

A current of 5 A exists in a square loop of side $\frac{1}{\sqrt{2}}\,\mathrm{m}$. Then the magnitude of the magnetic field B at the centre of the square loop will be $p \times 10^{-6}\,\mathrm{T}$. where, value of p is _____ [Take $\mu_0=4 \pi \times 10^{-7}\,\mathrm{T}\,\mathrm{mA}^{-1}$].

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Q73. Numerical

The least count of a screw gauge is 0.01 mm . If the pitch is increased by 75% and number of divisions on the circular scale is reduced by 50%, the new least count will be _____ $\times 10^{-3}\,\mathrm{mm}$

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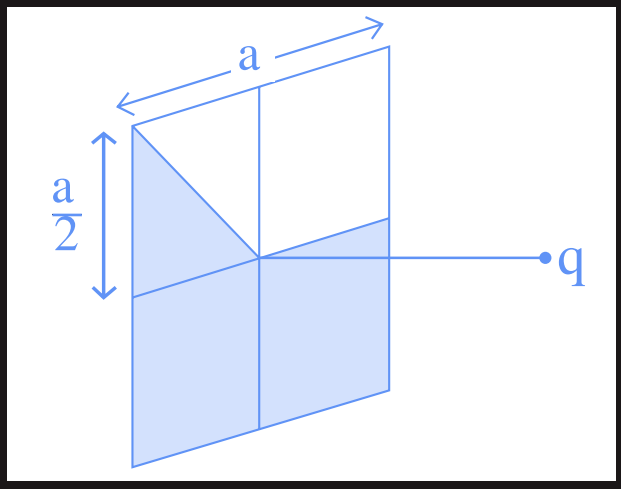
Q74. Numerical

The temperature of 1 mole of an ideal monoatomic gas is increased by $50^{\circ}\mathrm{C}$ at constant pressure. The total heat added and change in internal energy are E_1 and E_2, respectively. If $\frac{E_1}{E_2} = \frac{x}{9}$ then the value of x is _____.

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Q75. Numerical

A square loop of sides $a = 1\,\mathrm{m}$ is held normally in front of a point charge $\mathrm{q} = 1\mathrm{C}$ at a distance $\frac{\mathrm{a}}{2}$. The flux of the electric field through the shaded region is $\frac{5}{\mathrm{p}} \times \frac{1}{\varepsilon_0} \frac{\mathrm{Nm}^2}{\mathrm{C}}$, where the value of p is _____ .



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Answer Key

Chemistry

1. **C**	2. **A**	3. **A**	4. **C**	5. **D**
6. **C**	7. **C**	8. **C**	9. **C**	10. **A**
11. **A**	12. **A**	13. **C**	14. **D**	15. **B**
16. **C**	17. **D**	18. **A**	19. **A**	20. **C**
21. **962**	22. **4**	23. **420**	24. **61**	25. **700**

Mathematics

26. **D**	27. **D**	28. **D**	29. **B**	30. **C**
31. **D**	32. **A**	33. **C**	34. **A**	35. **D**
36. **C**	37. **B**	38. **A**	39. **B**	40. **C**
41. **D**	42. **B**	43. **A**	44. **C**	45. **C**
46. **44**	47. **19**	48. **5120**	49. **14**	50. **125**

Physics

51. **D**	52. **C**	53. **A**	54. **C**	55. **D**
56. **C**	57. **C**	58. **B**	59. **B**	60. **D**
61. **C**	62. **B**	63. **D**	64. **B**	65. **B**
66. **C**	67. **D**	68. **D**	69. **B**	70. **D**
71. **2**	72. **8**	73. **35**	74. **15**	75. **48**

For numerical answers given as a range, any value within the range is correct.

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